

Global CO₂ Emissions

Who Emits, How Much, and What's Changing

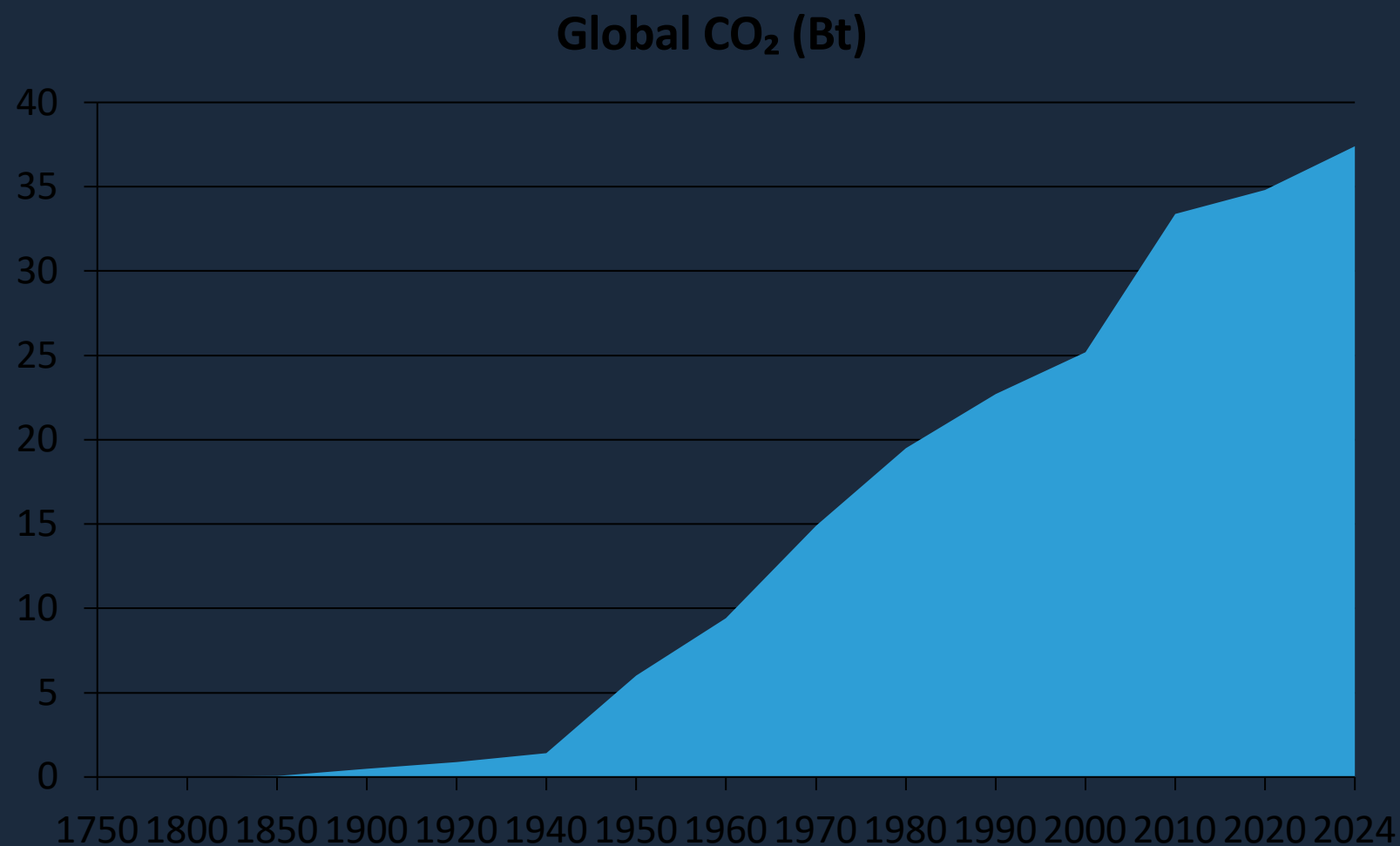
Fossil Fuels & Industry | 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data

Authors: Hannah Ritchie and Max Roser

For policy analysts, climate negotiators, and sustainability researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



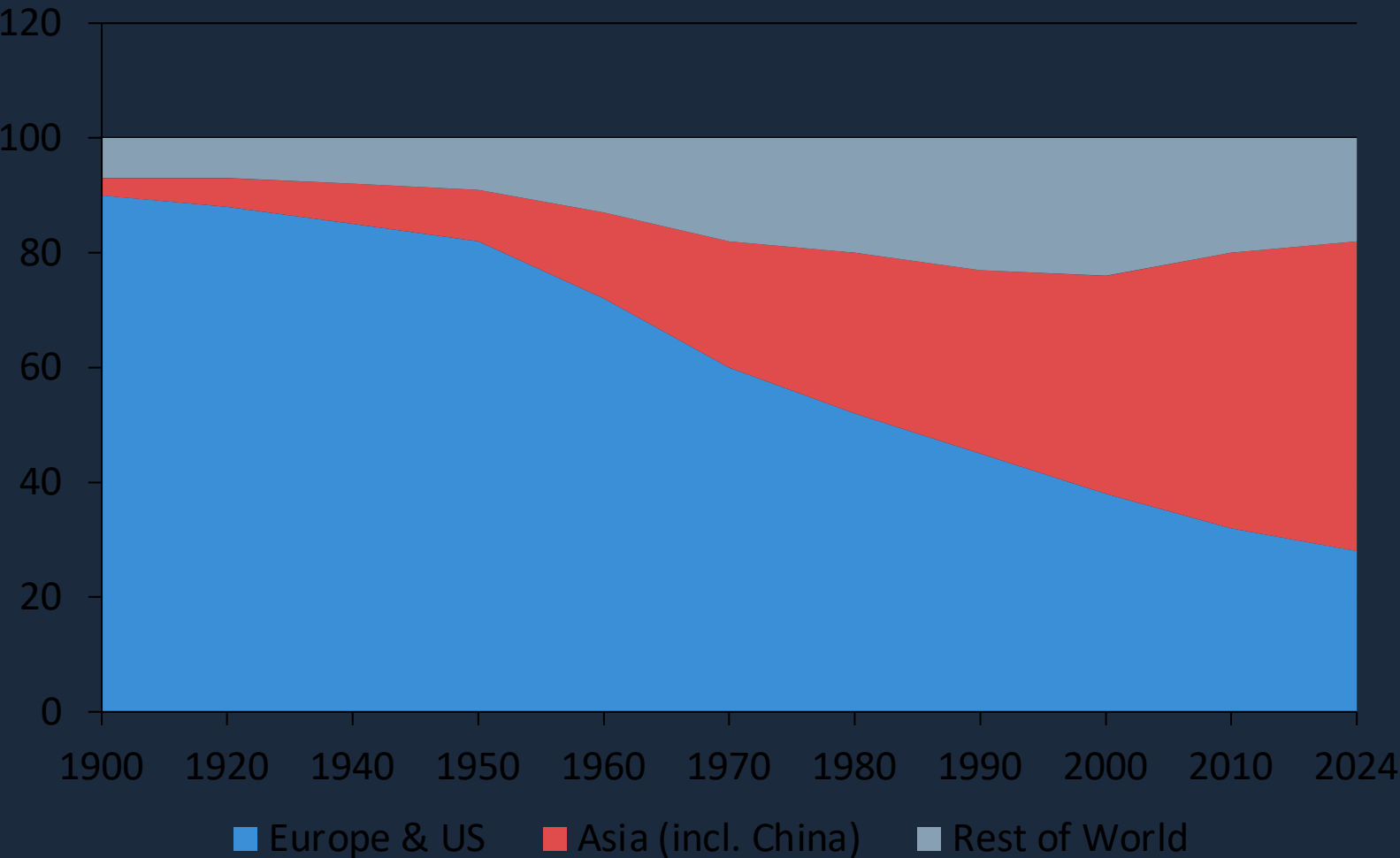
Key Inflection Points

- Pre-1950: Negligible**
Emissions under 6 Bt for two centuries of industrialization
- 1950–1990: Acceleration**
Emissions nearly quadrupled from 6 Bt to over 20 Bt
- 1990–today: Slowing but rising**
Growth rate decelerated; still no global peak

35+ Bt/yr
Global emissions in 2024
Still rising — no peak yet

Source: Global Carbon Budget 2025 via Our World in Data

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



The Shift in Numbers

1900

Europe & US: >90%
Asia: ~3%

1950

Europe & US: >85%
Asia: ~9%

2024

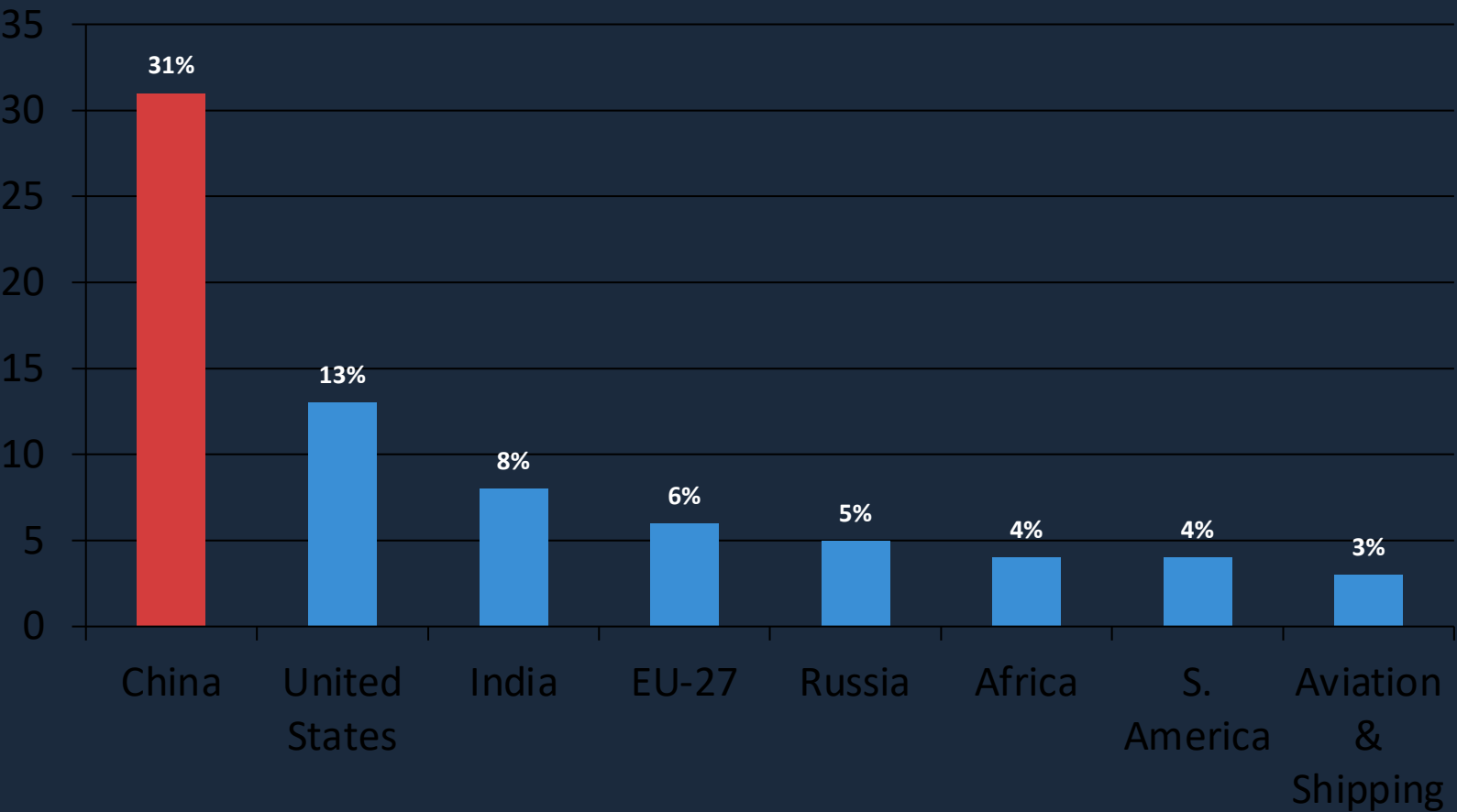
Europe & US: <30%
Asia: ~54%

Asia now produces roughly
half of global emissions
— led by China

Source: Global Carbon Budget 2025 via Our World in Data

China Now Emits More Than One-Quarter of Global CO₂

Share of Global CO₂ (%)

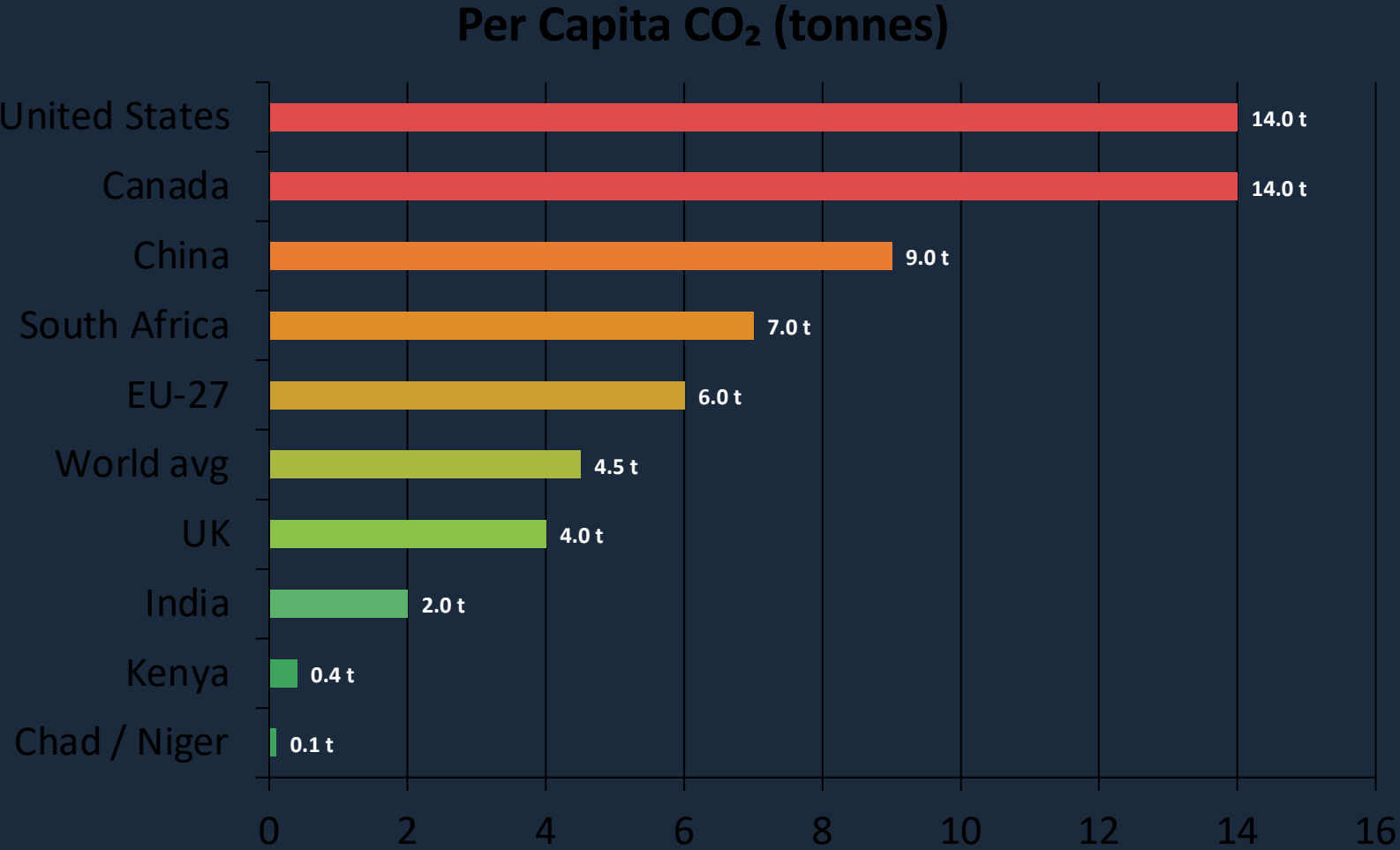


Country-Level Breakdown

- China: ~31% — largest annual emitter; growth slowing but still rising
- United States: ~13% — declining share
- India: ~8% — rising, still well below China
- EU-27: ~6% — declining
- Russia: ~5% — roughly stable
- Africa & S. America: 3–4% each — comparable to international aviation & shipping

Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



150× GAP

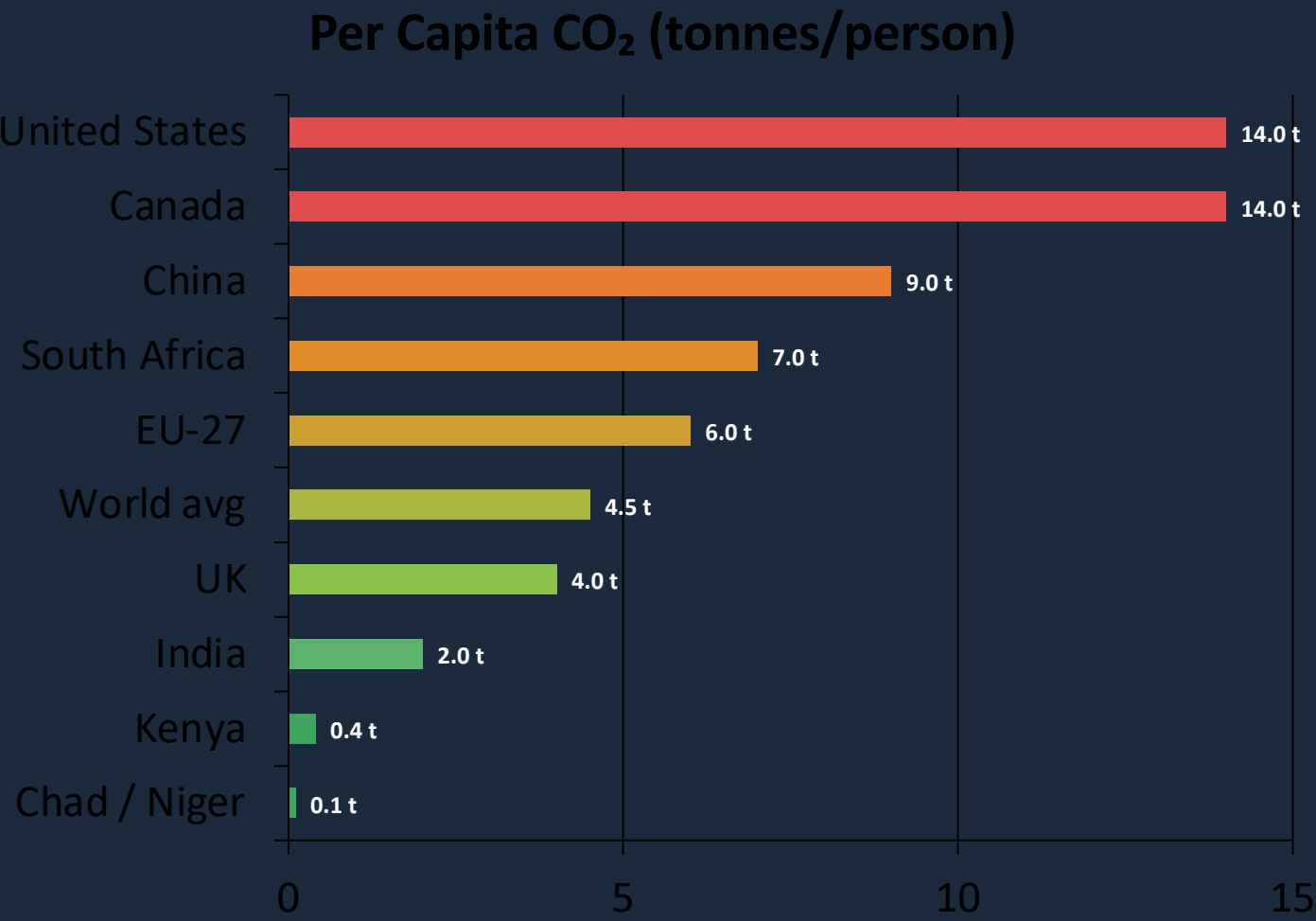
An American emits in under 2 days what a person in Chad or Niger emits in a year

Europe vs. US

France and the UK emit far less per capita than the US despite similar living standards

World average: ~4.5 t/person

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India



US vs. China

US per capita ~14 t | China ~9 t
China is the largest total emitter
but per capita remains well below the US

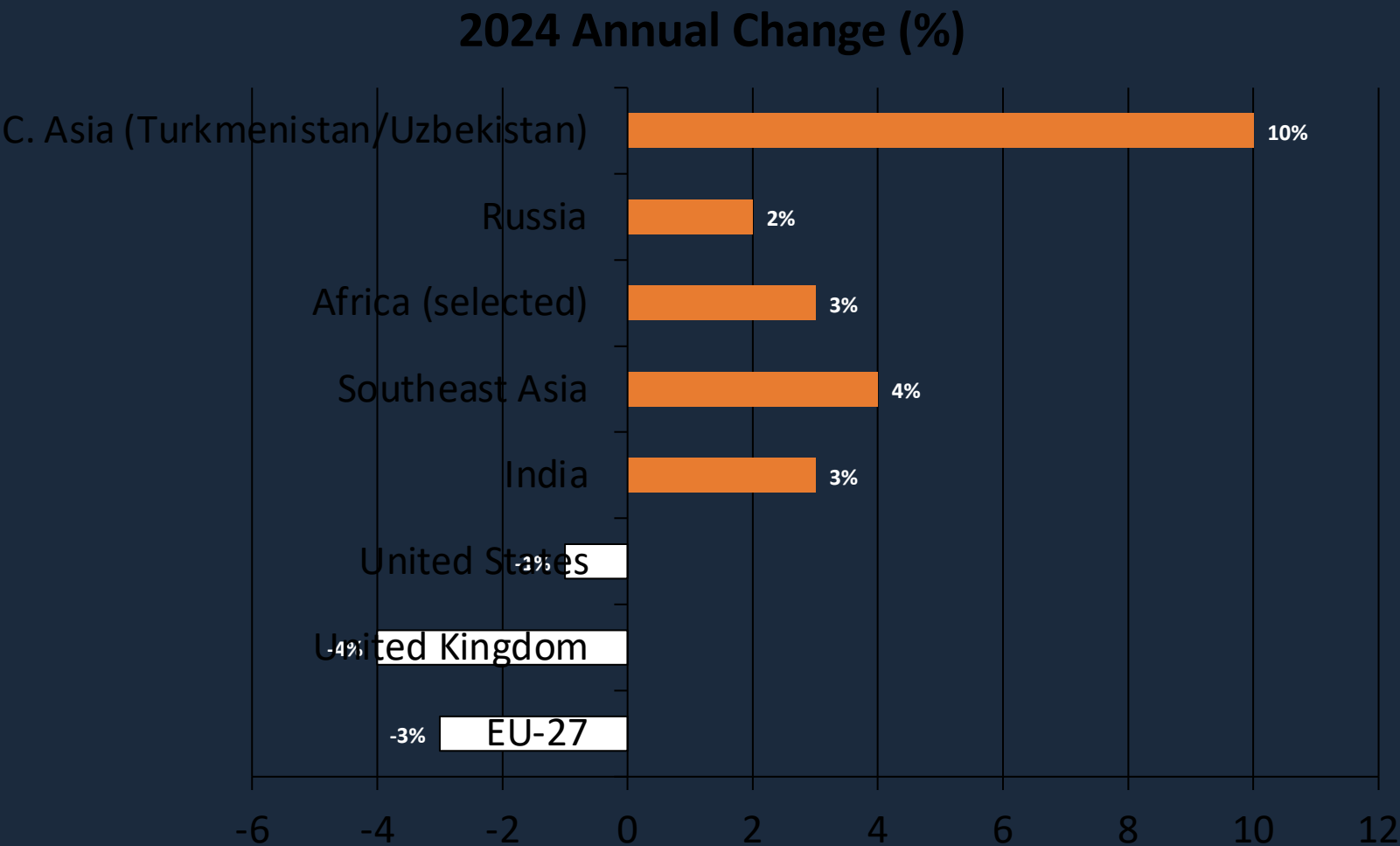
India: Still Very Low

Per capita ~2 t — less than half
the world average of ~4.5 t
Rapidly rising from a low base

Sub-Saharan Africa

Chad/Niger: ~0.1 t | Kenya: ~0.4 t
Entire continent accounts for
just 3–4% of global emissions

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



Declines (blue)

US, EU, UK saw emissions decline in 2024

Increases (orange)

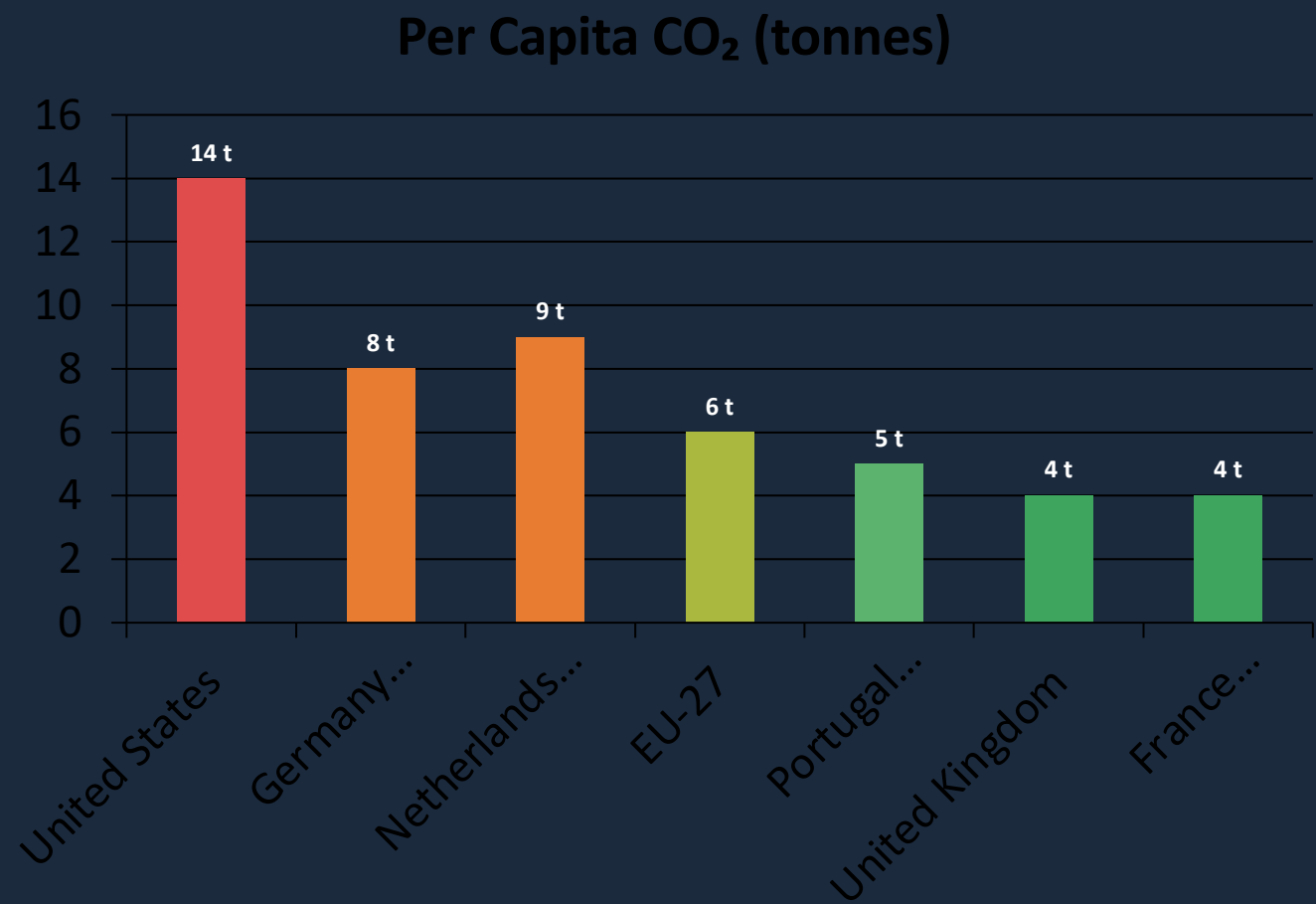
Russia, SE Asia, parts of Africa saw emissions rise

Notable Outlier

Central Asia (Turkmenistan/Uzbekistan) exceeded +10%

Source: Global Carbon Budget 2025 via Our World in Data | See figure_5.jpg for choropleth map

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



Why the Gap?

Among wealthy nations with similar prosperity levels, per capita emissions vary significantly because of energy mix choices.

France & the UK

Higher share of nuclear and renewable electricity → lower per capita emissions (~4 t)

Germany, Netherlands, US

Greater reliance on fossil fuels for electricity → higher per capita emissions (8–14 t)

Key Insight

Energy policy and technology choices can decouple prosperity from emissions.

Key Takeaways: Responsibility Is Shared but Unequal

1

No Peak Yet

Global CO₂ emissions from fossil fuels exceed 35 Bt/yr and have not yet peaked, despite slowing growth.

2

China: Largest but Not Richest

China is the largest annual emitter (~31%), but its per capita emissions (~9 t) remain well below the US (~14 t).

3

Historical Responsibility

Europe and the US dominated emissions historically (>90% in 1900) but now account for less than one-third.

4

150× Per Capita Gap

A 150× gap persists between the lowest emitters (Sub-Saharan Africa, ~0.1 t/person) and the highest (US/Canada/Australia, ~14–15 t/person).

5

Energy Policy Works

Countries with similar prosperity can have very different per capita emissions depending on their energy mix — France and the UK emit far less per capita than Germany or the US.